LTP Calculator: Reversal Price Formula & Calculation Rules

Extension of Support (EOS) aur Extension of Resistance (EOR) ka Technical Breakdown

1. Overview & Core Concept

LTP Calculator me **Reversal Price** (jaise *Extension of Support - EOS* ya *Extension of Resistance - EOR*) nikalne ka koi seedha manual static formula nahi hota hai.

Yeh software **Black-Scholes Options Pricing Model** aur **Options Greeks** (khas karke *Delta* aur *Implied Volatility - IV*) ke mathematical logic par kaam karta hai. Reversal Price basically woh Price Point (Index ya Stock Value) hota hai jahan par Option Buyer ka Intrinsic Value + Time Value aur Option Seller ke Risk Limit ka equilibrium/balance banta hai.

2. Core Mathematical Rule

CALL SIDE REVERSAL FORMULA (EOR / CALL REVERSAL PRICE)

$$\textit{Reversal Price} = \textit{Strike Price} + \textit{Call Premium} - (\textit{Time Decay Adjustments / Greek Effect})$$

PUT SIDE REVERSAL FORMULA (EOS / PUT REVERSAL PRICE)

$$\textit{Reversal Price} = \textit{Strike Price} - \textit{Put Premium} + (\textit{Time Decay Adjustments / Greek Effect})$$

3. Dynamic Adjustment Rule (Greeks & IV Effect)

Agar aap kewal $\textit{Strike Price} \pm \textit{Premium}$ karenge, toh woh sirf basic **Break-Even Level** banega, na ki real-time Reversal Price. Software isme live market data ke mutabiq 3 main factors adjust karta hai:

Factor	Mechanism / Formula
1. Delta-Adjusted Move	Index/Stock me kitna point move aane par premium kitna change hoga. $\Delta\ \textit{Premium} = \textit{Delta} \times \Delta\ \textit{Underlying Price}$

Factor	Mechanism / Formula
2. Implied Volatility (IV)	High IV hone par Reversal Level Strike Price se thoda aur dur (Extension) chala jayega. Low IV me Reversal level strike ke paas rehta hai.
3. Theoretical Price Equilibrium	LTP Calculator dekhta hai ki kis exact Index Level par us Strike Price ka Theoretical Call/Put Price ZERO ya Minimum Value ko touch karega.

4. Practical Live Example & Practical Application

Live Scenario Simulation

Market Data:

- **Nifty Spot:** 24,000
- **Support Strike Price:** 24,000 (Put Side)
- **24,000 PE Ka Current LTP (Premium):** ₹80
- **Implied Volatility (IV):** Normal (~14%)

Step-by-Step Calculation:

1. **Basic Break-Even Level:**
   $$24,000 - 80 = 23,920$$
2. **Live Adjustment (Greek & Theta Buffer):**
   Theta Decay aur IV fluctuation ke mutabiq ~₹12 ka intrinsic sensitivity add/adjust hota hai.
   $$EOS\ (Extension\ of\ Support) = 23,920 + 12 = 23,932$$

Testing Rule: Jab Nifty girte hue **23,932** ke paas aayega, tab 24,000 PE ke sellers maximum profit zone/ equilibrium par honge. Yahan se market ke bounce back (reverse) karne ke chances highest hote hain.

5. Reversal Calculate karne ke 3 Main Steps

1. **Upper Reversal (EOR):** Selected Strike Price me us strike ka Call Premium (LTP) ADD karein. Usme se 5–10% Volatility/Greek buffer minus karein.

2. **Lower Reversal (EOS):** Selected Strike Price me se us strike ka Put Premium (LTP) SUBTRACT karein. Usme 5–10% Volatility/Greek buffer add karein.

3. **Change of Reversal:** Jaise-jaise din beetega (Time Decay / Theta) aur Premium kam hoga, aapka Reversal Level bhi Strike Price ke aur kareeb shift hone lagega.